**DO NOW — RECALL**

# Lesson 148 · Network Hardware: Switches, Routers and

Name: _____ Class: _____ Date: _____

**1. What does a firewall generally use to permit or block a connection?**

**2. What data structure follows FIFO?**

**3. What is the result of an AND gate with inputs 1 and 0?**

**4. What is a foreign key used for?**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

A policy of rules, for example addresses, ports or connection state.

### 2. Model answer

A queue.

### 3. Model answer

0.

### 4. Model answer

Referencing a candidate key in another relation.

### By the end, you can…

• Identify the role of switches, routers, wireless access points, NICs and repeaters.

• Follow a frame within a LAN and a packet between LANs, distinguishing MAC and IP use.

• Select network hardware for a realistic school network and justify its placement.

### Vocabulary — know these terms

switch — Connects local devices and forwards data-link frames.

router — Forwards network-layer IP packets between networks.

access point — Connects wireless devices to a LAN.

NIC — Network interface controller for network connectivity.

NEW LEARNING

# Read, annotate and explain

### Local connections

A switch forwards Ethernet frames using learned MAC addresses.

### Across networks

Router consults IP destinations and routing information.

### Wireless access

Access point bridges wireless stations onto a network.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

**Two school laptops on one wired LAN share a file; one then accesses a website on another network. Trace the hardware.**

### Laptop and switch

Identify the facts and make an initial prediction.

### Router/default gateway

Explain the mechanism or trace the intermediate stage.

### Remote network

Within the LAN, NICs form frames and the switch forwards using MAC addresses. To reach another IP network, traffic is sent to a router/default gateway that forwards packets. The gateway rewrites the link framing per next hop.

**WE DO — GUIDED PRACTICE**

# Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Differentiate switch, router and access point with their forwarding roles. [6]

### Question 2

For a PC accessing a website, explain why both MAC and IP addresses are useful. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Design a network for two labs and a library with wireless devices and an internet connection. Label device placement and traffic path. [8]

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**A basic Ethernet switch chooses internet routes using destination website names.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.